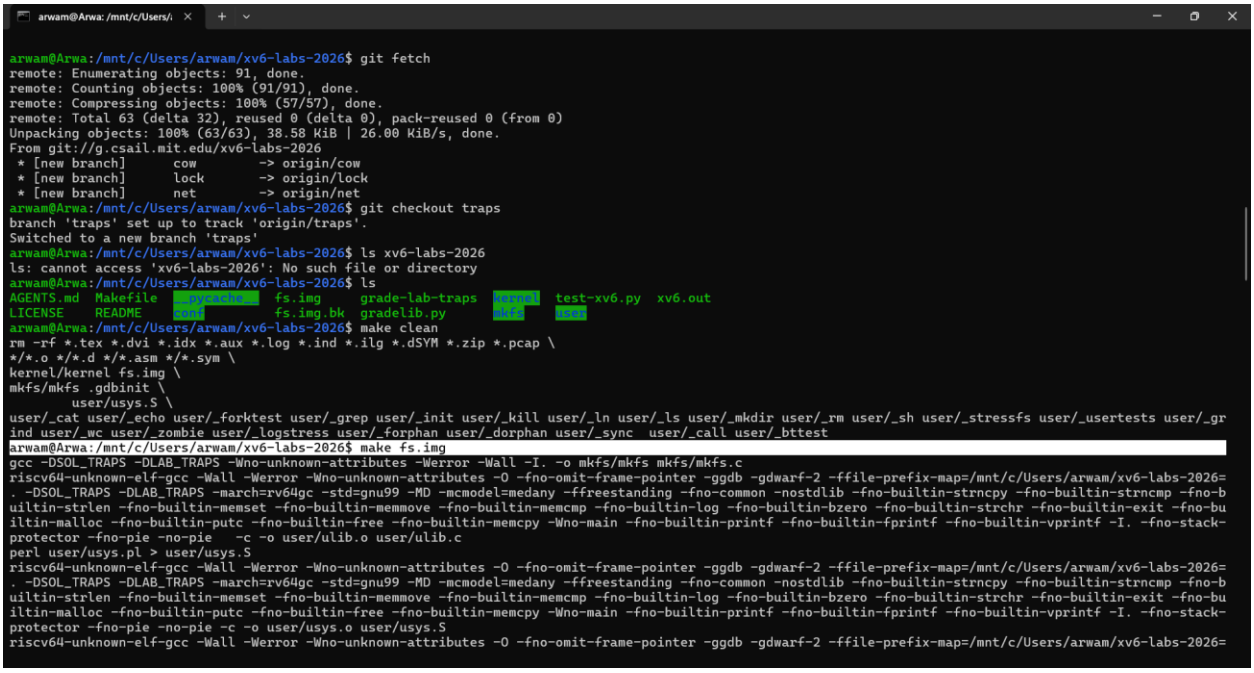
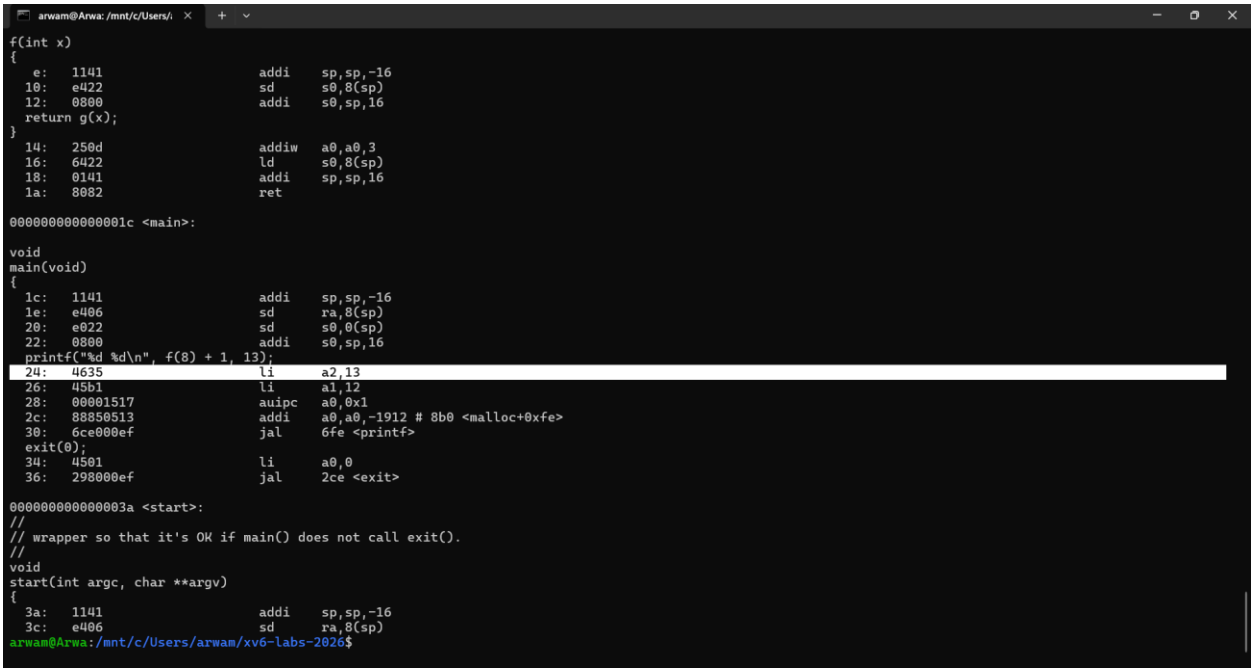
## RISC-V assembly (easy)



```
arwam@Arwa:/mnt/c/Users/arwam/xv6-labs-2026$ git fetch
remote: Enumerating objects: 91, done.
remote: Counting objects: 100% (91/91), done.
remote: Compressing objects: 100% (57/57), done.
remote: Total 63 (delta 32), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (63/63), 38.58 KiB | 26.00 KiB/s, done.
From git://g.csail.mit.edu/xv6-labs-2026
 * [new branch]      cow        -> origin/cow
 * [new branch]      lock       -> origin/lock
 * [new branch]      net        -> origin/net
arwam@Arwa:/mnt/c/Users/arwam/xv6-labs-2026$ git checkout traps
branch 'traps' set up to track 'origin/traps'.
Switched to a new branch 'traps'
arwam@Arwa:/mnt/c/Users/arwam/xv6-labs-2026$ ls xv6-labs-2026
ls: cannot access 'xv6-labs-2026': No such file or directory
arwam@Arwa:/mnt/c/Users/arwam/xv6-labs-2026$ ls
AGENTS.md  Makefile  __pycache__  fs.img     grade-lab-traps  kernel  test-xv6.py  xv6.out
LICENSE    README    conf         fs.img.bk  gradelib.py      mkfs    user
arwam@Arwa:/mnt/c/Users/arwam/xv6-labs-2026$ make clean
rm -rf *.tex *.dvi *.idx *.aux *.log *.ind *.ilg *.dSYM *.zip *.pcap \
*/*.o */*.d */*.asm */*.sym \
kernel/kernel fs.img \
mkfs/mkfs .gdbinit \
        user/usys.S \
user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_gr
ind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sync  user/_call user/_bttest
arwam@Arwa:/mnt/c/Users/arwam/xv6-labs-2026$ make fs.img
gcc -DSOL_TRAPS -DLAB_TRAPS -Wno-unknown-attributes -Werror -Wall -I. -o mkfs/mkfs mkfs/mkfs.c
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/mnt/c/Users/arwam/xv6-labs-2026=
. -DSOL_TRAPS -DLAB_TRAPS -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-b
uiltin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-bu
iltin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-
protector -fno-pie -no-pie   -c -o user/ulib.o user/ulib.c
perl user/usys.pl > user/usys.S
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/mnt/c/Users/arwam/xv6-labs-2026=
. -DSOL_TRAPS -DLAB_TRAPS -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-b
uiltin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-bu
iltin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-
protector -fno-pie -no-pie -c -o user/usys.o user/usys.S
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/mnt/c/Users/arwam/xv6-labs-2026=
```

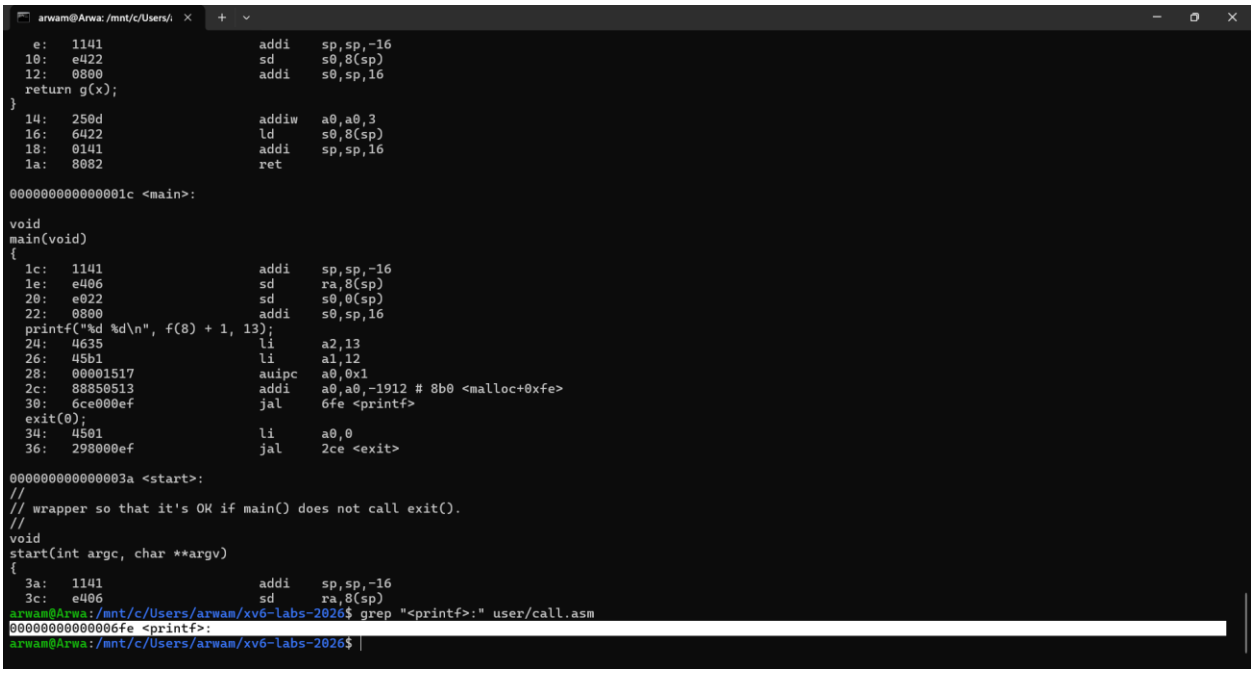


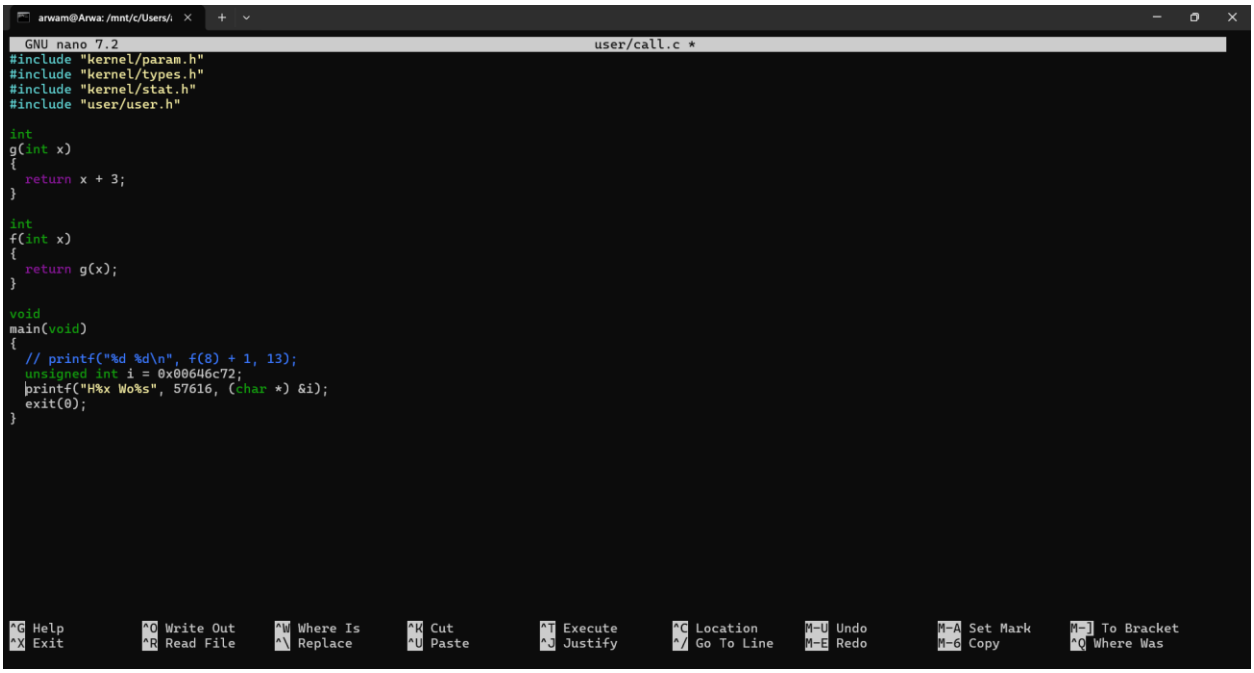
```
f(int x)
{
   e:   1141                    addi    sp,sp,-16
  10:   e422                    sd      s0,8(sp)
  12:   0800                    addi    s0,sp,16
  return g(x);
}
  14:   250d                    addiw   a0,a0,3
  16:   6422                    ld      s0,8(sp)
  18:   0141                    addi    sp,sp,16
  1a:   8082                    ret

000000000000001c <main>:

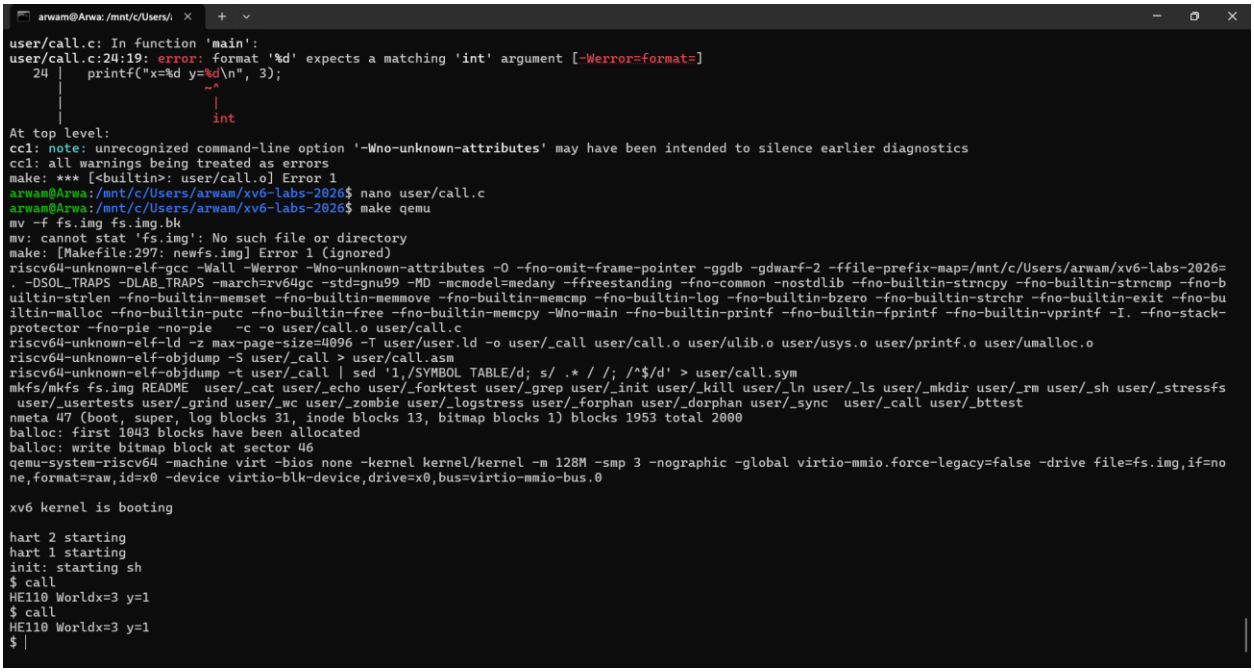
void
main(void)
{
  1c:   1141                    addi    sp,sp,-16
  1e:   e406                    sd      ra,8(sp)
  20:   e022                    sd      s0,0(sp)
  22:   0800                    addi    s0,sp,16
  printf("%d %d\n", f(8) + 1, 13);
  24:   4635                    li      a2,13
  26:   45b1                    li      a1,12
  28:   00001517                auipc   a0,0x1
  2c:   88850513                addi    a0,a0,-1912 # 8b0 <malloc+0xfe>
  30:   6ce000ef                jal     6fe <printf>
  exit(0);
  34:   4501                    li      a0,0
  36:   298000ef                jal     2ce <exit>

000000000000003a <start>:
//
// wrapper so that it's OK if main() does not call exit().
//
void
start(int argc, char **argv)
{
  3a:   1141                    addi    sp,sp,-16
  3c:   e406                    sd      ra,8(sp)
arwam@Arwa:/mnt/c/Users/arwam/xv6-labs-2026$
```

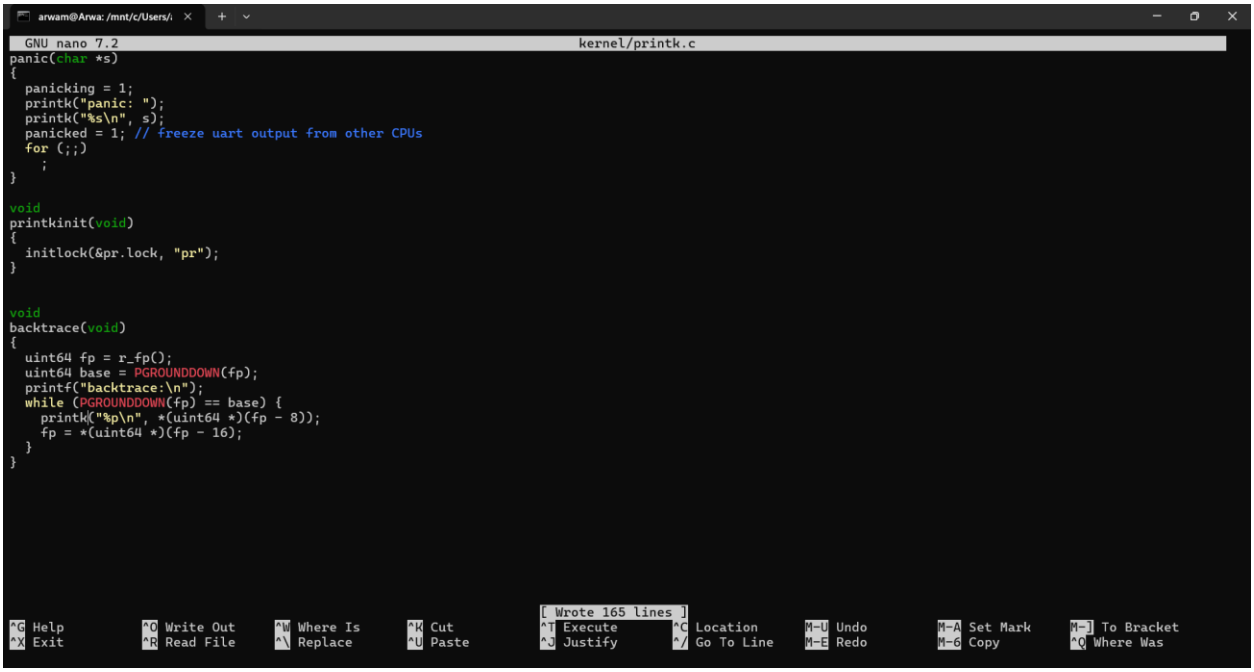

arwam@Arwa: /mnt/c/Users/a
e: 1141 addi sp,sp,-16
10: e422 sd s0,8(sp)
12: 0800 addi s0,sp,16
return g(x);
}
14: 250d addiw a0,a0,3
16: 6422 ld s0,8(sp)
18: 0141 addi sp,sp,16
1a: 8082 ret
000000000000001c <main>:
void
main(void)
{
1c: 1141 addi sp,sp,-16
1e: e406 sd ra,8(sp)
20: e022 sd s0,0(sp)
22: 0800 addi s0,sp,16
printf("%d %d\n", f(8) + 1, 13);
24: 4635 li a2,13
26: 45b1 li a1,12
28: 00001517 auipc a0,0x1
2c: 88850513 addi a0,a0,-1912 # 8b0 <malloc+0xfe>
30: 6ce000ef jal 6fe <printf>
exit(0);
34: 4501 li a0,0
36: 298000ef jal 2ce <exit>
000000000000003a <start>:
//
// wrapper so that it's OK if main() does not call exit().
//
void
start(int argc, char **argv)
{
3a: 1141 addi sp,sp,-16
3c: e406 sd ra,8(sp)
arwam@Arwa:/mnt/c/Users/arwam/xv6-labs-2026$ grep "<printf>:" user/call.asm
00000000000006fe <printf>:
arwam@Arwa:/mnt/c/Users/arwam/xv6-labs-2026$

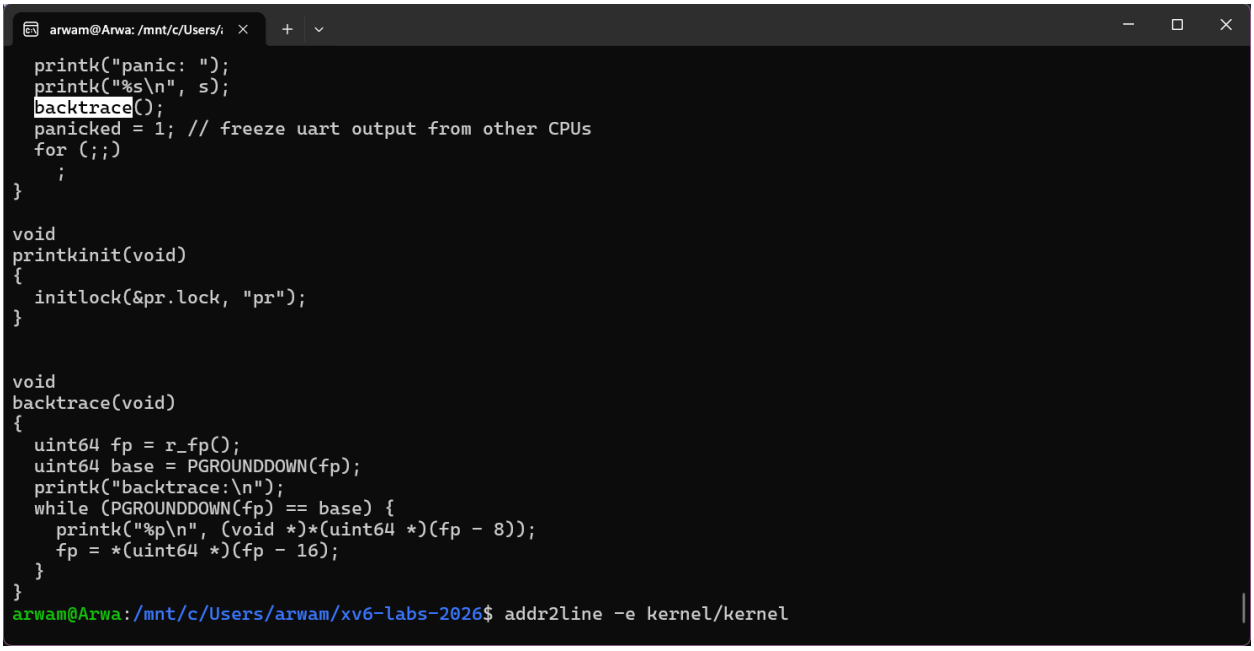

arwam@Arwa: /mnt/c/Users/a
GNU nano 7.2 user/call.c *
#include "kernel/param.h"
#include "kernel/types.h"
#include "kernel/stat.h"
#include "user/user.h"
int
g(int x)
{
return x + 3;
}
int
f(int x)
{
return g(x);
}
void
main(void)
{
// printf("%d %d\n", f(8) + 1, 13);
unsigned int i = 0x00646c72;
printf("H%x Wo%s", 57616, (char *) &i);
exit(0);
}
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was

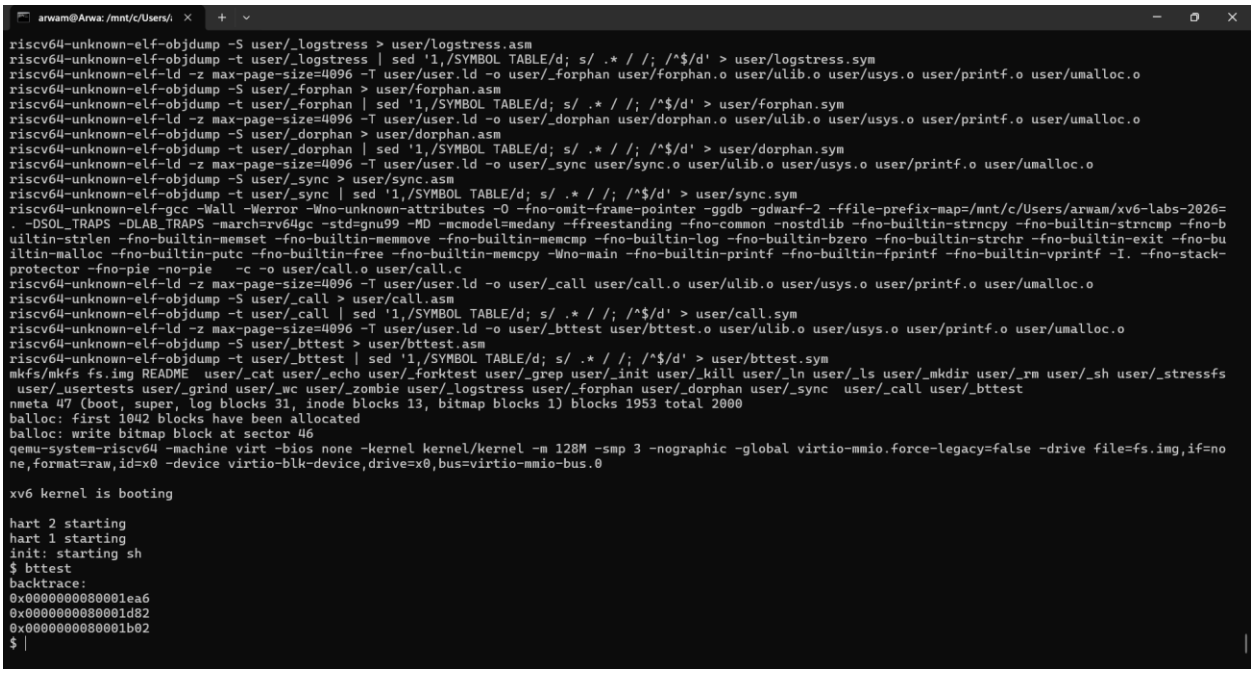

arwam@Arwa: /mnt/c/Users/a
protector -fno-pie -no-pie  -c -o kernel/printk.o kernel/printk.c
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/mnt/c/Users/arwam/xv6-labs-2026=. -DSOL_TRAPS -DLAB_TRAPS -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie  -c -o kernel/uart.o kernel/uart.c
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/mnt/c/Users/arwam/xv6-labs-2026=. -DSOL_TRAPS -DLAB_TRAPS -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie  -c -o kernel/spinlock.o kernel/spinlock.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T kernel/kernel.ld -o kernel/kernel kernel/entry.o kernel/kalloc.o kernel/string.o kernel/main.o kernel/vm.o kernel/proc.o kernel/swtch.o kernel/trampoline.o kernel/trap.o kernel/syscall.o kernel/sysproc.o kernel/bio.o kernel/fs.o kernel/log.o kernel/sleeplock.o kernel/file.o kernel/pipe.o kernel/exec.o kernel/sysfile.o kernel/kernelvec.o kernel/plic.o kernel/virtio_disk.o kernel/start.o kernel/console.o kernel/printk.o kernel/uart.o kernel/spinlock.o
riscv64-unknown-elf-ld: warning: kernel/kernel has a LOAD segment with RWX permissions
riscv64-unknown-elf-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-unknown-elf-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/mnt/c/Users/arwam/xv6-labs-2026=. -DSOL_TRAPS -DLAB_TRAPS -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie   -c -o user/call.o user/call.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_call user/call.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_call > user/call.asm
riscv64-unknown-elf-objdump -t user/_call | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/call.sym
mkfs/mkfs fs.img README  user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sync  user/_call user/_bttest
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
balloc: first 1042 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0
xv6 kernel is booting
hart 2 starting
hart 1 starting
init: starting sh
$ call
HE110 World$

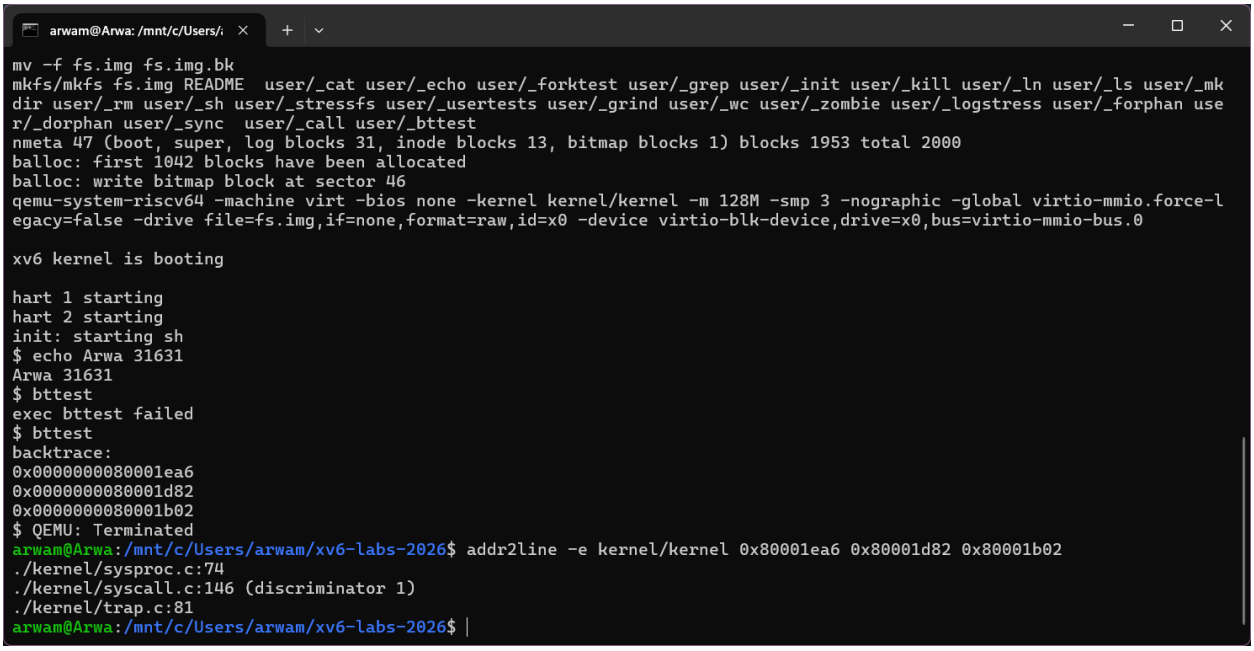

arwam@Arwa: /mnt/c/Users/a
user/call.c: In function 'main':
user/call.c:24:19: error: format '%d' expects a matching 'int' argument [-Werror=format=]
   24 |   printf("x=%d y=%d\n", 3);
      |                  ~^
      |                   |
      |                   int
At top level:
cc1: note: unrecognized command-line option '-Wno-unknown-attributes' may have been intended to silence earlier diagnostics
cc1: all warnings being treated as errors
make: *** [<builtin>: user/call.o] Error 1
arwam@Arwa:/mnt/c/Users/arwam/xv6-labs-2026$ nano user/call.c
arwam@Arwa:/mnt/c/Users/arwam/xv6-labs-2026$ make qemu
mv -f fs.img fs.img.bk
mv: cannot stat 'fs.img': No such file or directory
make: [Makefile:297: newfs.img] Error 1 (ignored)
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/mnt/c/Users/arwam/xv6-labs-2026=. -DSOL_TRAPS -DLAB_TRAPS -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie   -c -o user/call.o user/call.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_call user/call.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_call > user/call.asm
riscv64-unknown-elf-objdump -t user/_call | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/call.sym
mkfs/mkfs fs.img README  user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sync  user/_call user/_bttest
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
balloc: first 1043 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0
xv6 kernel is booting
hart 2 starting
hart 1 starting
init: starting sh
$ call
HE110 Worldx=3 y=1
$ call
HE110 Worldx=3 y=1
$

## Backtrace

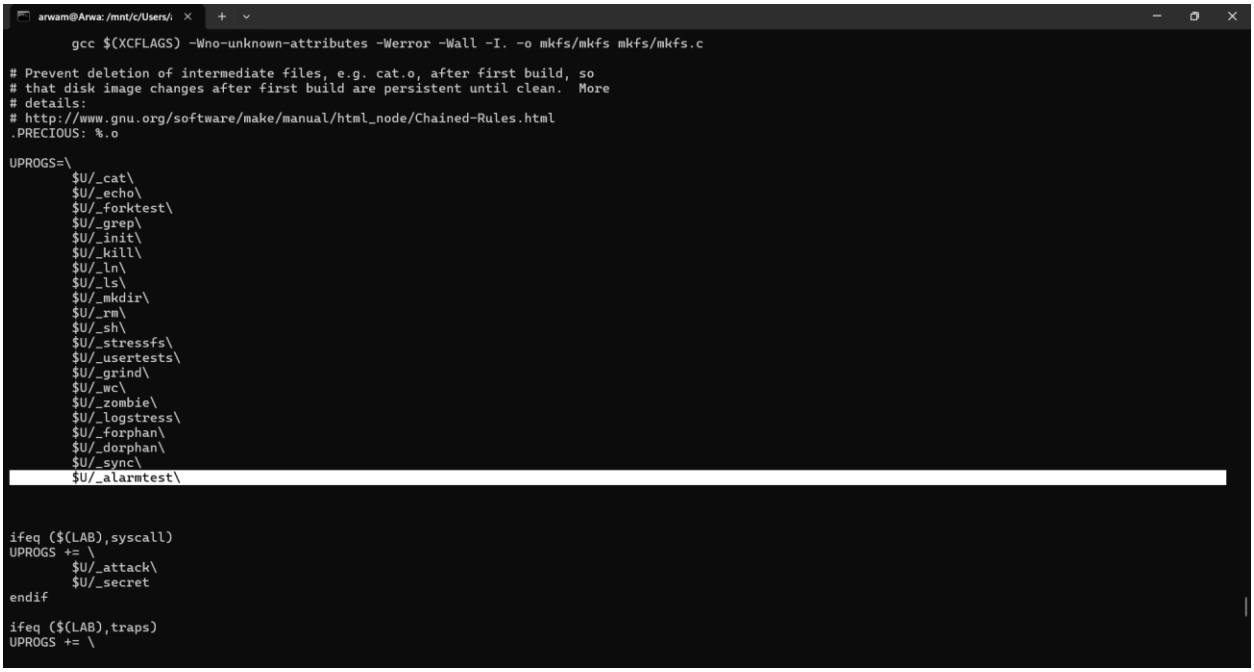

arwam@Arwa: /mnt/c/Users/
GNU nano 7.2
kernel/printk.c
panic(char *s)
{
panicking = 1;
printk("panic: ");
printk("%s\n", s);
panicked = 1; // freeze uart output from other CPUs
for (;;)
;
}
void
printkinit(void)
{
initlock(&pr.lock, "pr");
}
void
backtrace(void)
{
uint64 fp = r_fp();
uint64 base = PGROUNDDOWN(fp);
printf("backtrace:\n");
while (PGROUNDDOWN(fp) == base) {
printk("%p\n", *(uint64 *)(fp - 8));
fp = *(uint64 *)(fp - 16);
}
}
[ Wrote 165 lines ]
^G Help
^O Write Out
^W Where Is
^K Cut
^T Execute
^C Location
M-U Undo
M-A Set Mark
M-] To Bracket
^X Exit
^R Read File
^\ Replace
^U Paste
^J Justify
^/ Go To Line
M-E Redo
M-6 Copy
^Q Where Was


arwam@Arwa: /mnt/c/Users/
printk("panic: ");
printk("%s\n", s);
backtrace();
panicked = 1; // freeze uart output from other CPUs
for (;;)
;
}
void
printkinit(void)
{
initlock(&pr.lock, "pr");
}
void
backtrace(void)
{
uint64 fp = r_fp();
uint64 base = PGROUNDDOWN(fp);
printk("backtrace:\n");
while (PGROUNDDOWN(fp) == base) {
printk("%p\n", (void *)*(uint64 *)(fp - 8));
fp = *(uint64 *)(fp - 16);
}
}
arwam@Arwa:/mnt/c/Users/arwam/xv6-labs-2026$ addr2line -e kernel/kernel


arwam@Arwa: /mnt/c/Users/a
riscv64-unknown-elf-objdump -S user/_logstress > user/logstress.asm
riscv64-unknown-elf-objdump -t user/_logstress | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/logstress.sym
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_forphan user/forphan.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_forphan > user/forphan.asm
riscv64-unknown-elf-objdump -t user/_forphan | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/forphan.sym
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_dorphan user/dorphan.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_dorphan > user/dorphan.asm
riscv64-unknown-elf-objdump -t user/_dorphan | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/dorphan.sym
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_sync user/sync.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_sync > user/sync.asm
riscv64-unknown-elf-objdump -t user/_sync | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/sync.sym
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/mnt/c/Users/arwam/xv6-labs-2026=. -DSOL_TRAPS -DLAB_TRAPS -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie   -c -o user/call.o user/call.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_call user/call.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_call > user/call.asm
riscv64-unknown-elf-objdump -t user/_call | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/call.sym
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_bttest user/bttest.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_bttest > user/bttest.asm
riscv64-unknown-elf-objdump -t user/_bttest | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/bttest.sym
mkfs/mkfs fs.img README  user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sync  user/_call user/_bttest
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
balloc: first 1042 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0
xv6 kernel is booting
hart 2 starting
hart 1 starting
init: starting sh
$ bttest
backtrace:
0x0000000080001ea6
0x0000000080001d82
0x0000000080001b02
$


arwam@Arwa: /mnt/c/Users/a
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README  user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sync  user/_call user/_bttest
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
balloc: first 1042 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0
xv6 kernel is booting
hart 1 starting
hart 2 starting
init: starting sh
$ echo Arwa 31631
Arwa 31631
$ bttest
exec bttest failed
$ bttest
backtrace:
0x0000000080001ea6
0x0000000080001d82
0x0000000080001b02
$ QEMU: Terminated
arwam@Arwa:/mnt/c/Users/arwam/xv6-labs-2026$ addr2line -e kernel/kernel 0x80001ea6 0x80001d82 0x80001b02
./kernel/sysproc.c:74
./kernel/syscall.c:146 (discriminator 1)
./kernel/trap.c:81
arwam@Arwa:/mnt/c/Users/arwam/xv6-labs-2026$

## Alarm (Hard)



```
arwam@Arwa: /mnt/c/Users/
        gcc $(XCFLAGS) -Wno-unknown-attributes -Werror -Wall -I. -o mkfs/mkfs mkfs/mkfs.c

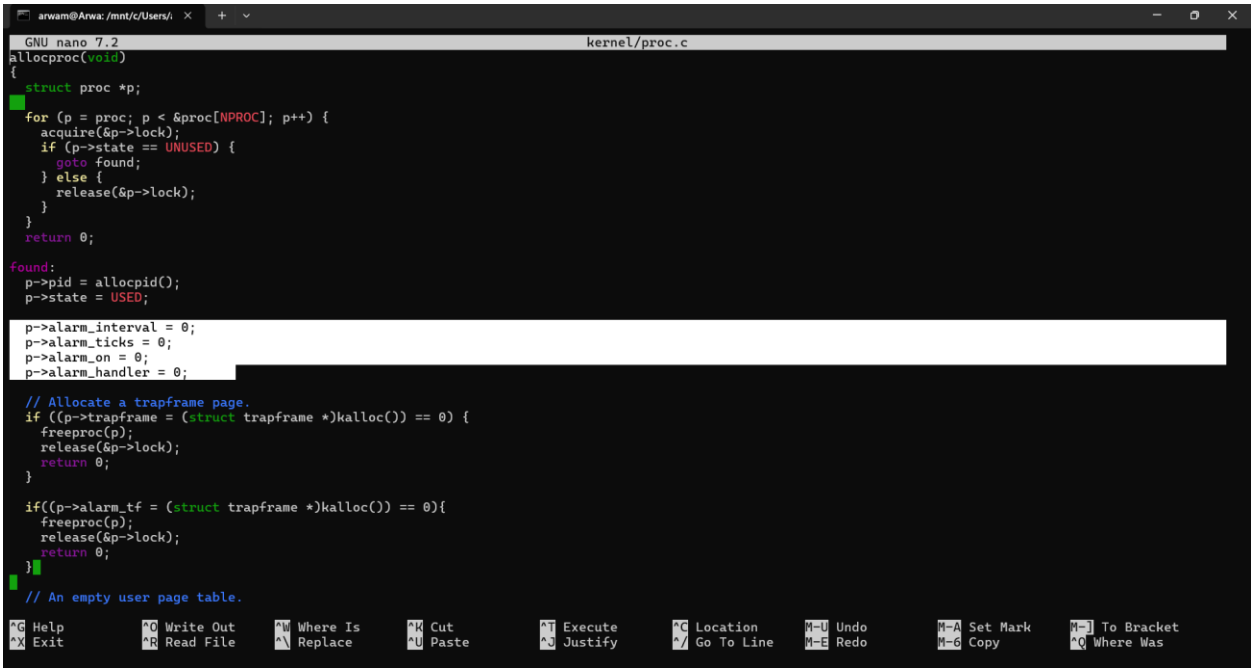
# Prevent deletion of intermediate files, e.g. cat.o, after first build, so
# that disk image changes after first build are persistent until clean.  More
# details:
# http://www.gnu.org/software/make/manual/html_node/Chained-Rules.html
.PRECIOUS: %.o

UPROGS=\
        $U/_cat\
        $U/_echo\
        $U/_forktest\
        $U/_grep\
        $U/_init\
        $U/_kill\
        $U/_ln\
        $U/_ls\
        $U/_mkdir\
        $U/_rm\
        $U/_sh\
        $U/_stressfs\
        $U/_usertests\
        $U/_grind\
        $U/_wc\
        $U/_zombie\
        $U/_logstress\
        $U/_forphan\
        $U/_dorphan\
        $U/_sync\
        $U/_alarmtest\



ifeq ($(LAB),syscall)
UPROGS += \
        $U/_attack\
        $U/_secret
endif

ifeq ($(LAB),traps)
UPROGS += \
```



```
arwam@Arwa: /mnt/c/Users/
 GNU nano 7.2                                                kernel/proc.c
allocproc(void)
{
  struct proc *p;

  for (p = proc; p < &proc[NPROC]; p++) {
    acquire(&p->lock);
    if (p->state == UNUSED) {
      goto found;
    } else {
      release(&p->lock);
    }
  }
  return 0;

found:
  p->pid = allocpid();
  p->state = USED;

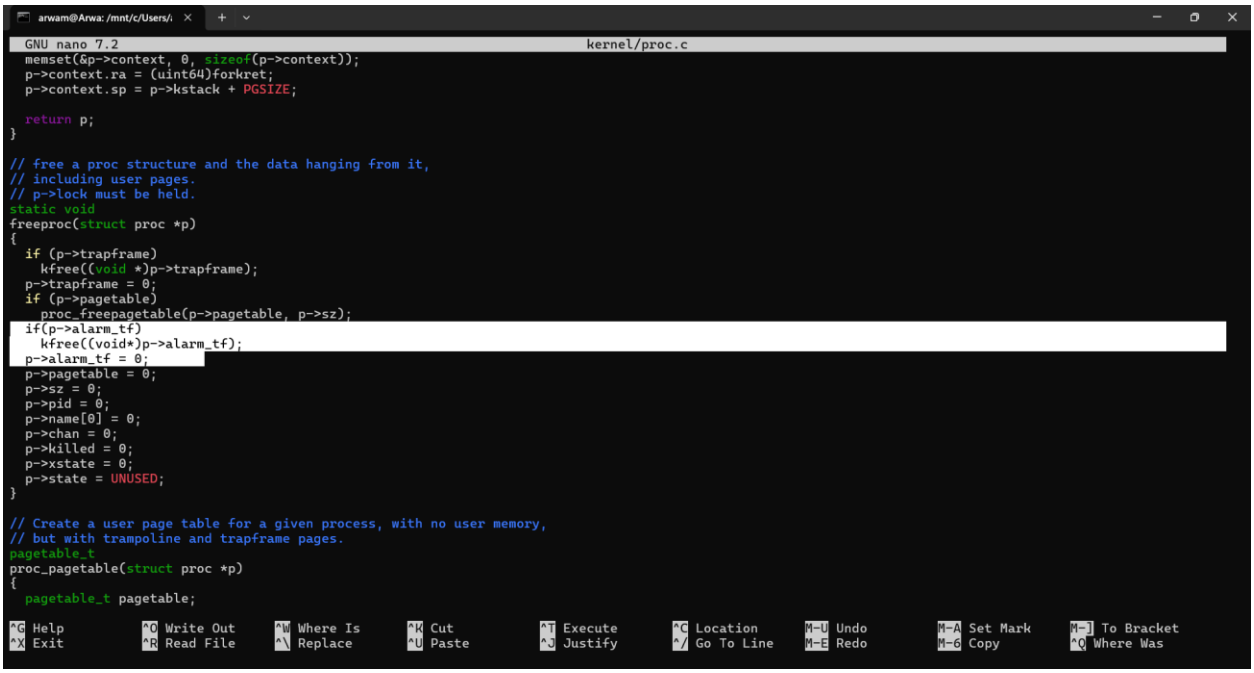
  p->alarm_interval = 0;
  p->alarm_ticks = 0;
  p->alarm_on = 0;
  p->alarm_handler = 0;

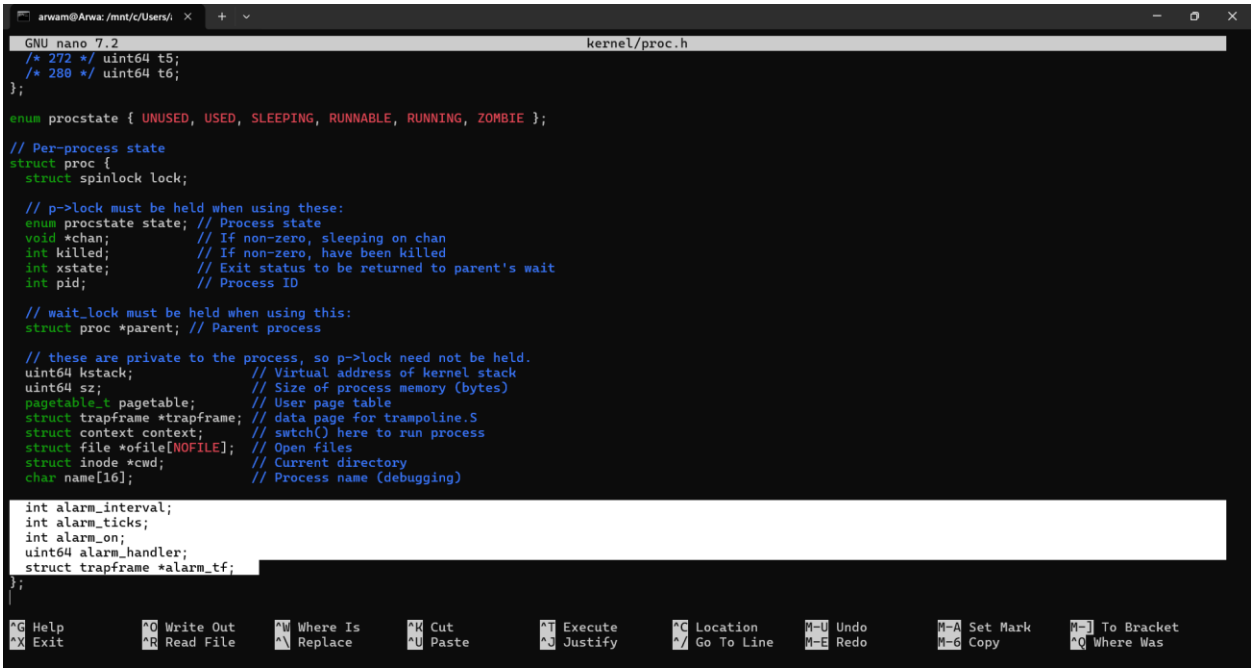
  // Allocate a trapframe page.
  if ((p->trapframe = (struct trapframe *)kalloc()) == 0) {
    freeproc(p);
    release(&p->lock);
    return 0;
  }

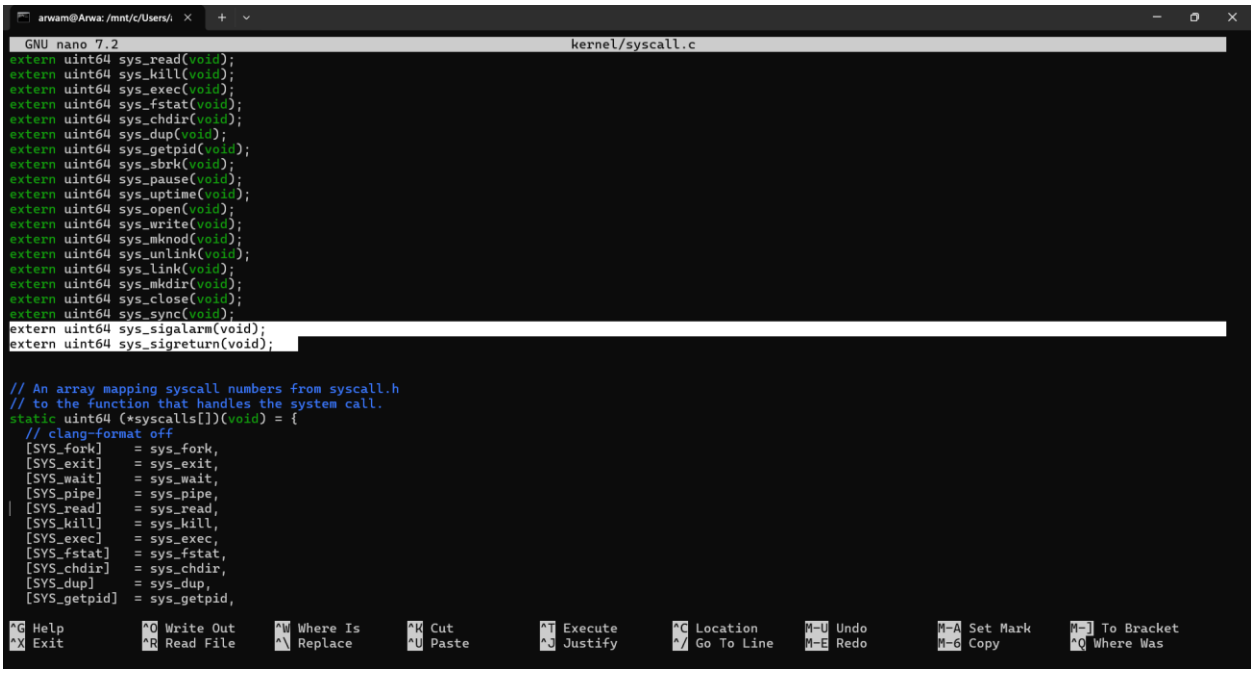
  if((p->alarm_tf = (struct trapframe *)kalloc()) == 0){
    freeproc(p);
    release(&p->lock);
    return 0;
  }

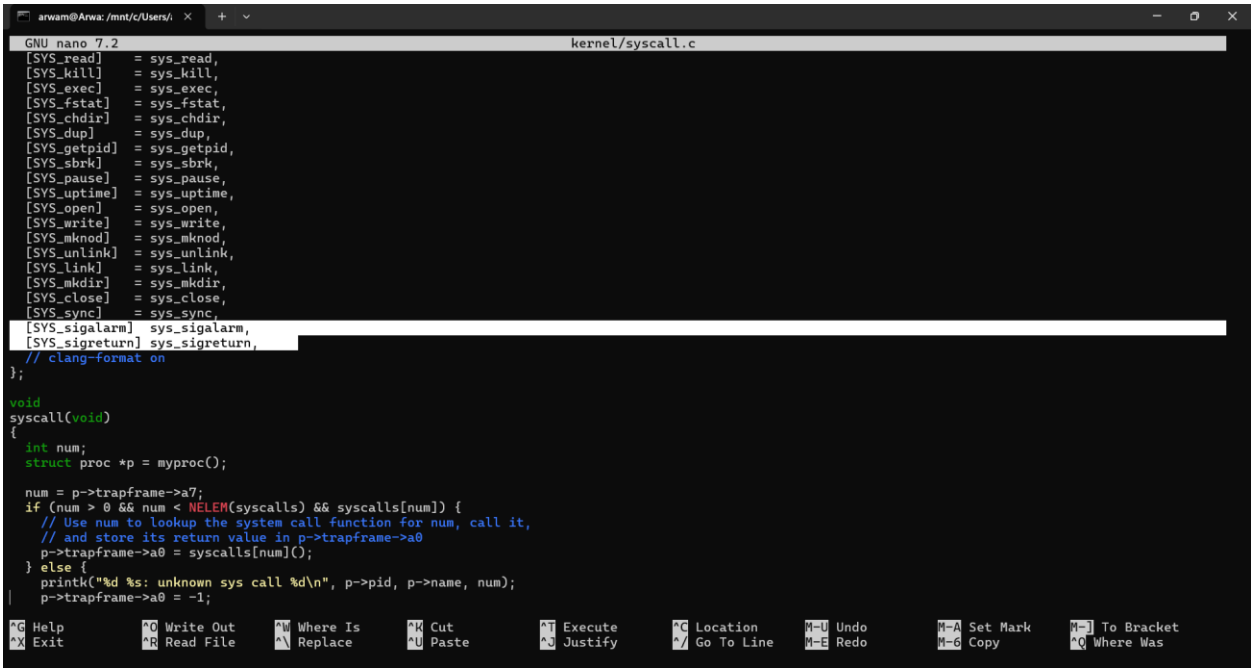
  // An empty user page table.

^G Help        ^O Write Out   ^W Where Is   ^K Cut         ^T Execute     ^C Location    M-U Undo       M-A Set Mark   M-] To Bracket
^X Exit        ^R Read File   ^\ Replace    ^U Paste       ^J Justify     ^/ Go To Line  M-E Redo       M-6 Copy       ^Q Where Was
```

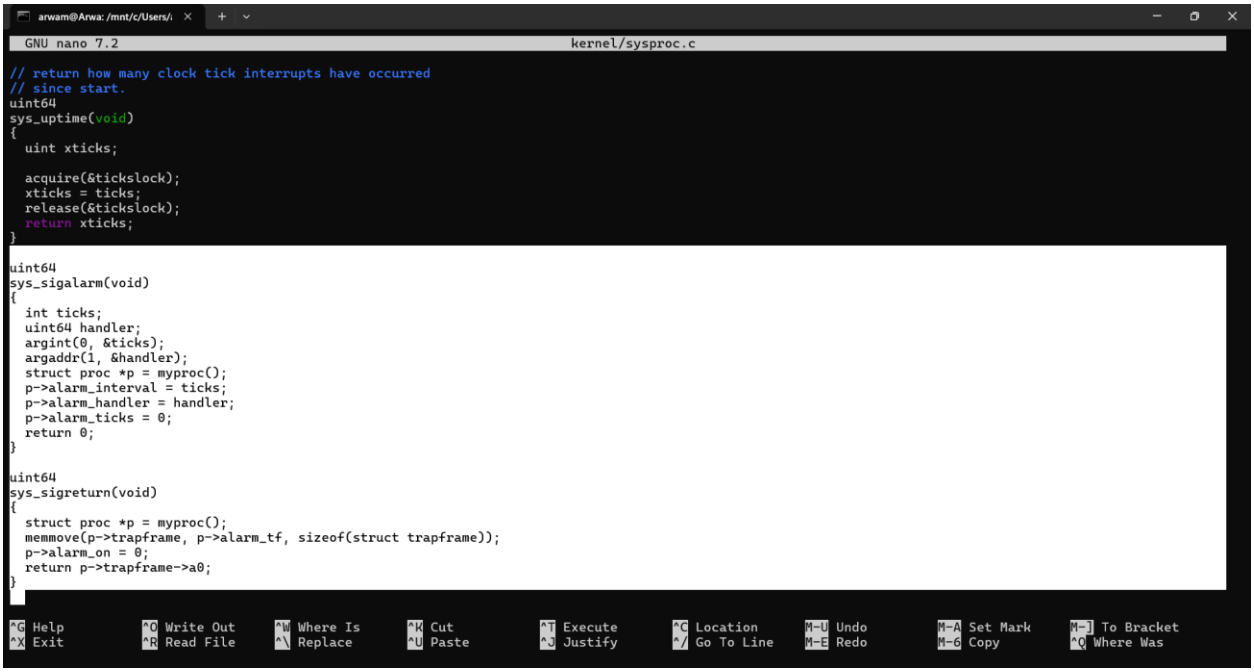

arwam@Arwa: /mnt/c/Users/
GNU nano 7.2 kernel/proc.c
memset(&p->context, 0, sizeof(p->context));
p->context.ra = (uint64)forkret;
p->context.sp = p->kstack + PGSIZE;
return p;
}
// free a proc structure and the data hanging from it,
// including user pages.
// p->lock must be held.
static void
freeproc(struct proc *p)
{
if (p->trapframe)
kfree((void *)p->trapframe);
p->trapframe = 0;
if (p->pagetable)
proc_freepagetable(p->pagetable, p->sz);
if(p->alarm_tf)
kfree((void*)p->alarm_tf);
p->alarm_tf = 0;
p->pagetable = 0;
p->sz = 0;
p->pid = 0;
p->name[0] = 0;
p->chan = 0;
p->killed = 0;
p->xstate = 0;
p->state = UNUSED;
}
// Create a user page table for a given process, with no user memory,
// but with trampoline and trapframe pages.
pagetable_t
proc_pagetable(struct proc *p)
{
pagetable_t pagetable;
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was

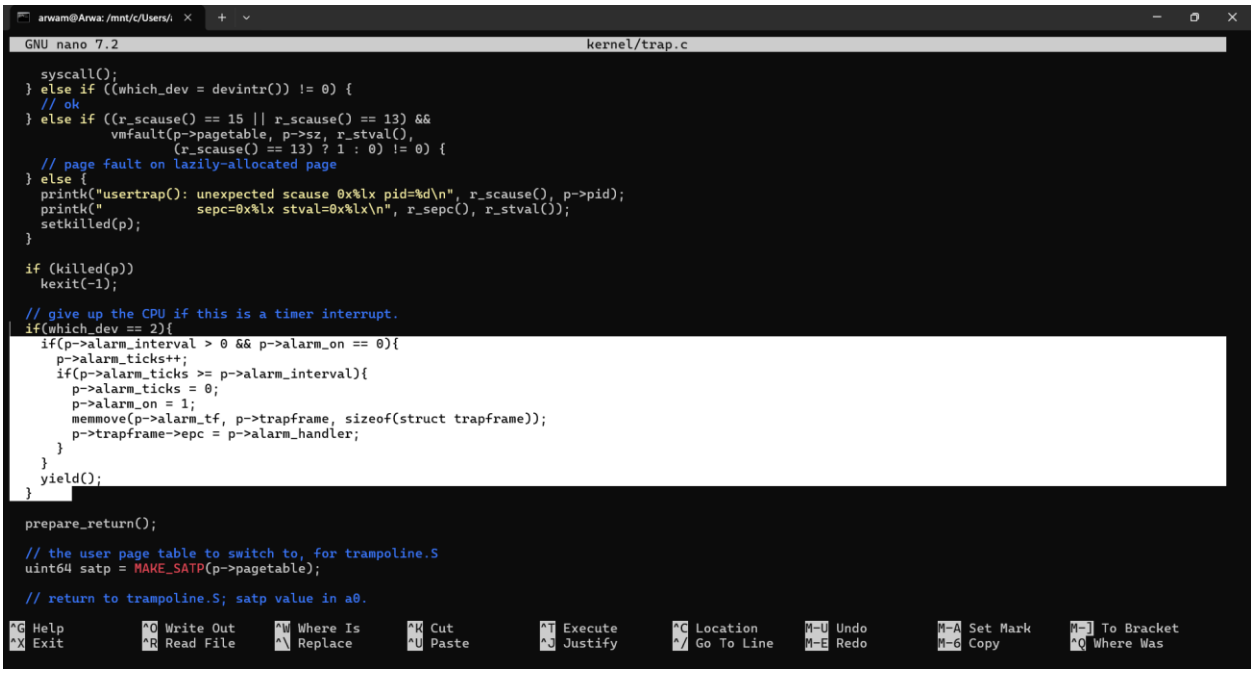

arwam@Arwa: /mnt/c/Users/
GNU nano 7.2 kernel/proc.h
/* 272 */ uint64 t5;
/* 280 */ uint64 t6;
};
enum procstate { UNUSED, USED, SLEEPING, RUNNABLE, RUNNING, ZOMBIE };
// Per-process state
struct proc {
struct spinlock lock;
// p->lock must be held when using these:
enum procstate state; // Process state
void *chan; // If non-zero, sleeping on chan
int killed; // If non-zero, have been killed
int xstate; // Exit status to be returned to parent's wait
int pid; // Process ID
// wait_lock must be held when using this:
struct proc *parent; // Parent process
// these are private to the process, so p->lock need not be held.
uint64 kstack; // Virtual address of kernel stack
uint64 sz; // Size of process memory (bytes)
pagetable_t pagetable; // User page table
struct trapframe *trapframe; // data page for trampoline.S
struct context context; // swtch() here to run process
struct file *ofile[NOFILE]; // Open files
struct inode *cwd; // Current directory
char name[16]; // Process name (debugging)
int alarm_interval;
int alarm_ticks;
int alarm_on;
uint64 alarm_handler;
struct trapframe *alarm_tf;
};
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was

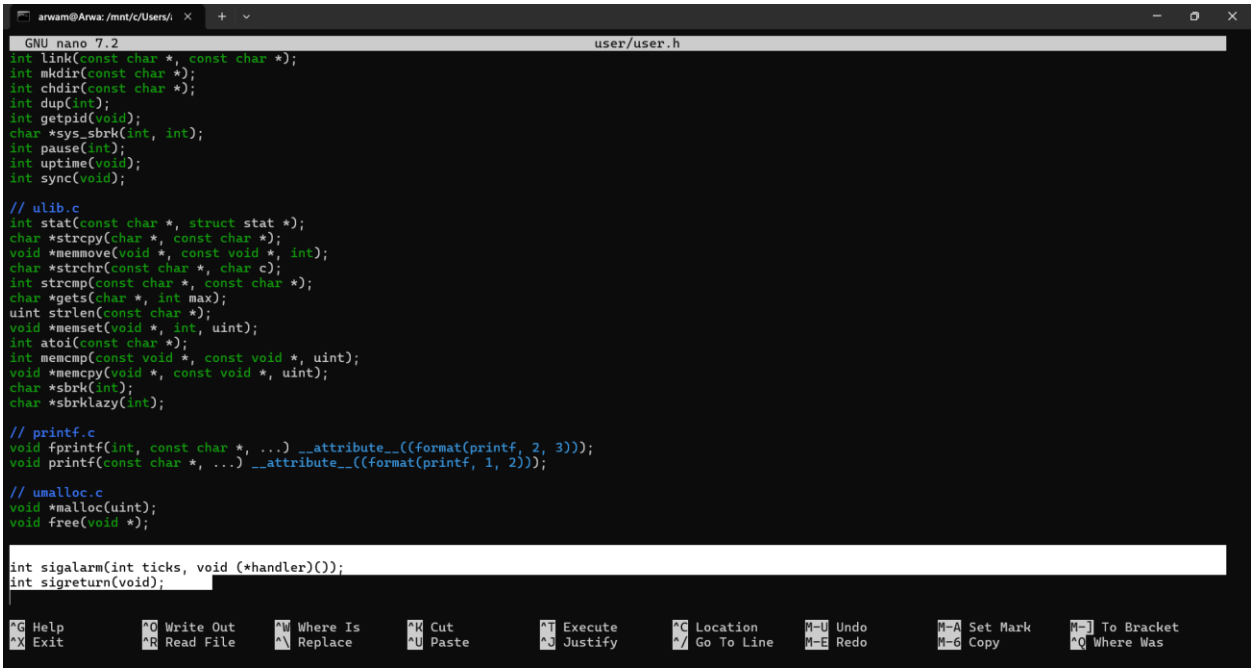

arwam@Arwa: /mnt/c/Users/
GNU nano 7.2 kernel/syscall.c
extern uint64 sys_read(void);
extern uint64 sys_kill(void);
extern uint64 sys_exec(void);
extern uint64 sys_fstat(void);
extern uint64 sys_chdir(void);
extern uint64 sys_dup(void);
extern uint64 sys_getpid(void);
extern uint64 sys_sbrk(void);
extern uint64 sys_pause(void);
extern uint64 sys_uptime(void);
extern uint64 sys_open(void);
extern uint64 sys_write(void);
extern uint64 sys_mknod(void);
extern uint64 sys_unlink(void);
extern uint64 sys_link(void);
extern uint64 sys_mkdir(void);
extern uint64 sys_close(void);
extern uint64 sys_sync(void);
extern uint64 sys_sigalarm(void);
extern uint64 sys_sigreturn(void);
// An array mapping syscall numbers from syscall.h
// to the function that handles the system call.
static uint64 (*syscalls[])(void) = {
// clang-format off
[SYS_fork] = sys_fork,
[SYS_exit] = sys_exit,
[SYS_wait] = sys_wait,
[SYS_pipe] = sys_pipe,
[SYS_read] = sys_read,
[SYS_kill] = sys_kill,
[SYS_exec] = sys_exec,
[SYS_fstat] = sys_fstat,
[SYS_chdir] = sys_chdir,
[SYS_dup] = sys_dup,
[SYS_getpid] = sys_getpid,
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was

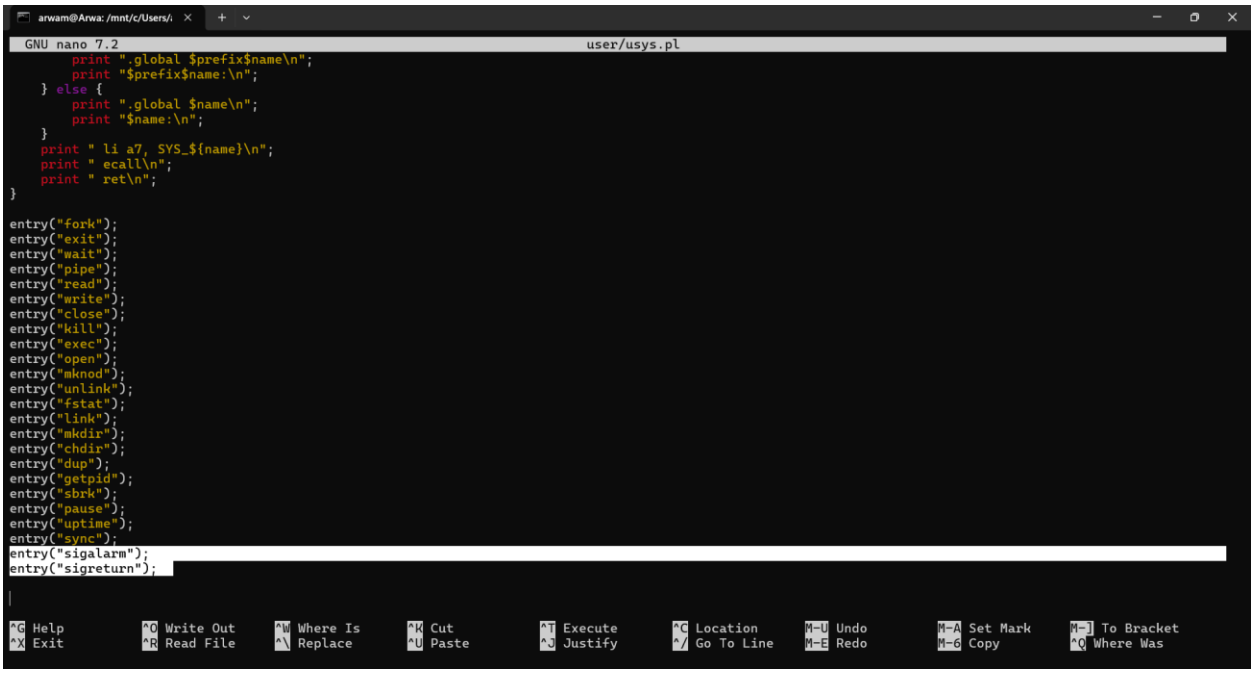

arwam@Arwa: /mnt/c/Users/
GNU nano 7.2 kernel/syscall.c
[SYS_read] = sys_read,
[SYS_kill] = sys_kill,
[SYS_exec] = sys_exec,
[SYS_fstat] = sys_fstat,
[SYS_chdir] = sys_chdir,
[SYS_dup] = sys_dup,
[SYS_getpid] = sys_getpid,
[SYS_sbrk] = sys_sbrk,
[SYS_pause] = sys_pause,
[SYS_uptime] = sys_uptime,
[SYS_open] = sys_open,
[SYS_write] = sys_write,
[SYS_mknod] = sys_mknod,
[SYS_unlink] = sys_unlink,
[SYS_link] = sys_link,
[SYS_mkdir] = sys_mkdir,
[SYS_close] = sys_close,
[SYS_sync] = sys_sync,
[SYS_sigalarm] sys_sigalarm,
[SYS_sigreturn] sys_sigreturn,
// clang-format on
};
void
syscall(void)
{
int num;
struct proc *p = myproc();
num = p->trapframe->a7;
if (num > 0 && num < NELEM(syscalls) && syscalls[num]) {
// Use num to lookup the system call function for num, call it,
// and store its return value in p->trapframe->a0
p->trapframe->a0 = syscalls[num]();
} else {
printk("%d %s: unknown sys call %d\n", p->pid, p->name, num);
p->trapframe->a0 = -1;
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was

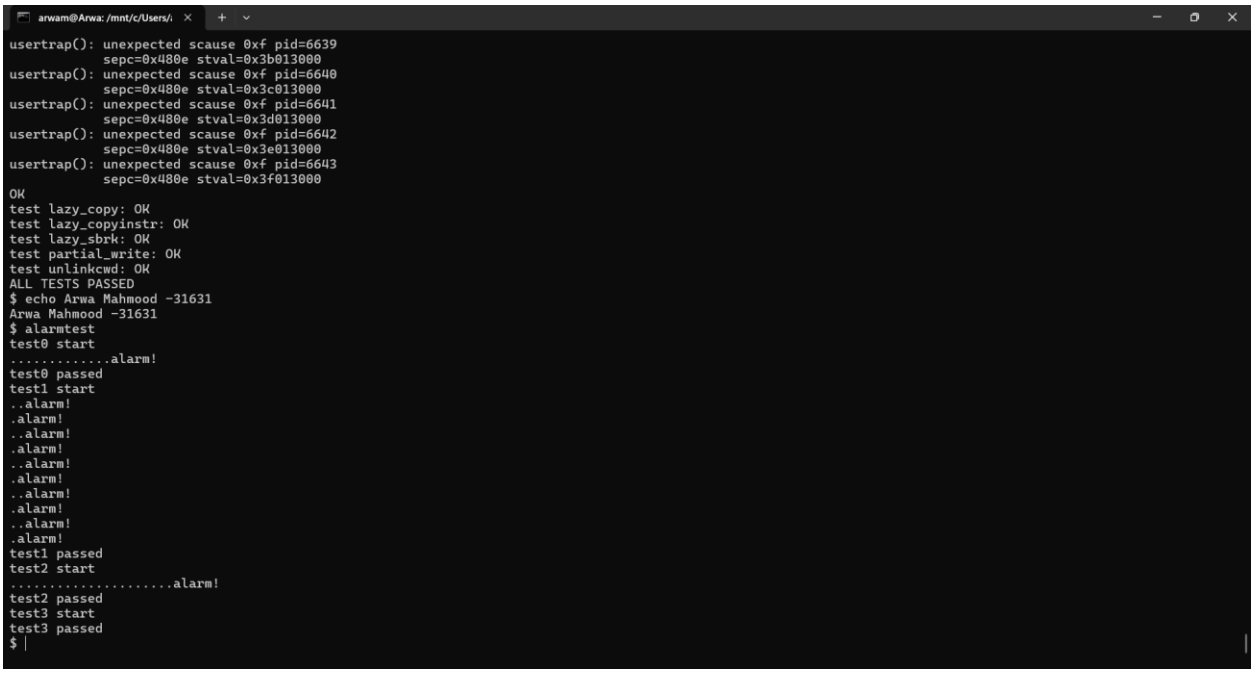

arwam@Arwa: /mnt/c/Users/
GNU nano 7.2
kernel/syscall.h
// System call numbers
#define SYS_fork 1
#define SYS_exit 2
#define SYS_wait 3
#define SYS_pipe 4
#define SYS_read 5
#define SYS_kill 6
#define SYS_exec 7
#define SYS_fstat 8
#define SYS_chdir 9
#define SYS_dup 10
#define SYS_getpid 11
#define SYS_sbrk 12
#define SYS_pause 13
#define SYS_uptime 14
#define SYS_open 15
#define SYS_write 16
#define SYS_mknod 17
#define SYS_unlink 18
#define SYS_link 19
#define SYS_mkdir 20
#define SYS_close 21
#define SYS_sync 22
#define SYS_sigalarm 23
#define SYS_sigreturn 24
[ Read 25 lines ]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was

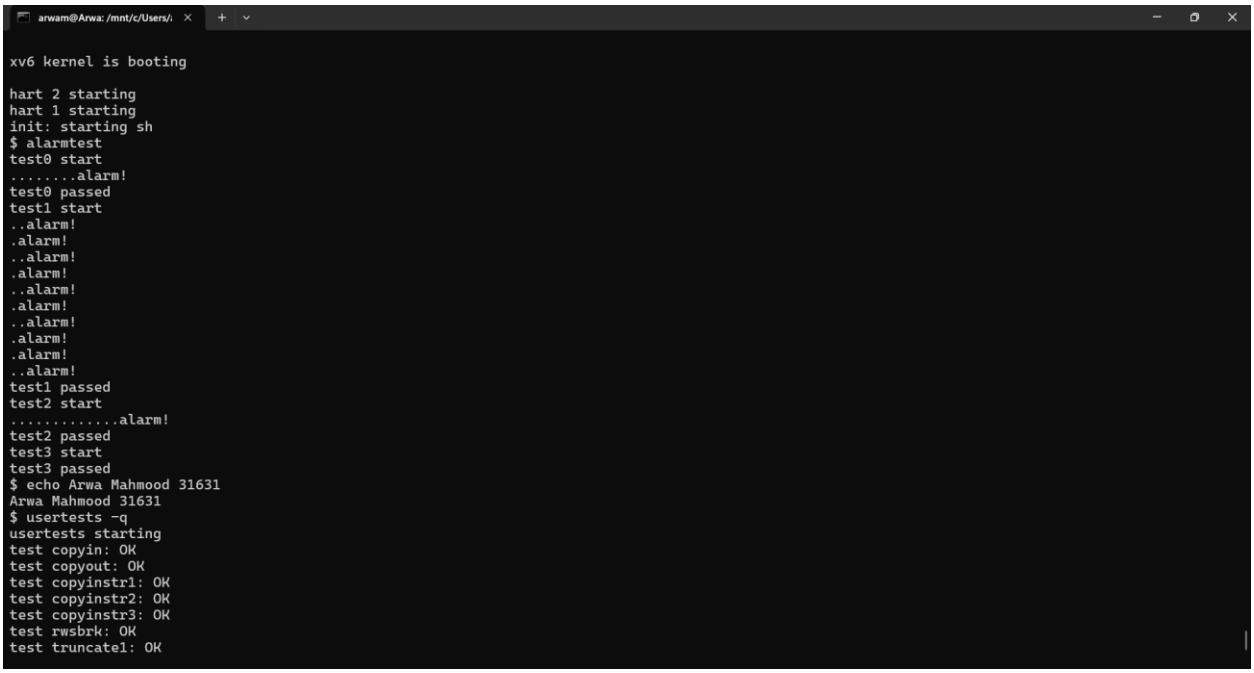

arwam@Arwa: /mnt/c/Users/
GNU nano 7.2
kernel/sysproc.c
// return how many clock tick interrupts have occurred
// since start.
uint64
sys_uptime(void)
{
uint xticks;
acquire(&tickslock);
xticks = ticks;
release(&tickslock);
return xticks;
}
uint64
sys_sigalarm(void)
{
int ticks;
uint64 handler;
argint(0, &ticks);
argaddr(1, &handler);
struct proc *p = myproc();
p->alarm_interval = ticks;
p->alarm_handler = handler;
p->alarm_ticks = 0;
return 0;
}
uint64
sys_sigreturn(void)
{
struct proc *p = myproc();
memmove(p->trapframe, p->alarm_tf, sizeof(struct trapframe));
p->alarm_on = 0;
return p->trapframe->a0;
}
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was


arwam@Arwa: /mnt/c/Users/
GNU nano 7.2
kernel/trap.c
syscall();
} else if ((which_dev = devintr()) != 0) {
// ok
} else if ((r_scause() == 15 || r_scause() == 13) &&
vmfault(p->pagetable, p->sz, r_stval(),
(r_scause() == 13) ? 1 : 0) != 0) {
// page fault on lazily-allocated page
} else {
printk("usertrap(): unexpected scause 0x%lx pid=%d\n", r_scause(), p->pid);
printk("sepc=0x%lx stval=0x%lx\n", r_sepc(), r_stval());
setkilled(p);
}
if (killed(p))
kexit(-1);
// give up the CPU if this is a timer interrupt.
if(which_dev == 2){
if(p->alarm_interval > 0 && p->alarm_on == 0){
p->alarm_ticks++;
if(p->alarm_ticks >= p->alarm_interval){
p->alarm_ticks = 0;
p->alarm_on = 1;
memmove(p->alarm_tf, p->trapframe, sizeof(struct trapframe));
p->trapframe->epc = p->alarm_handler;
}
}
yield();
}
prepare_return();
// the user page table to switch to, for trampoline.S
uint64 satp = MAKE_SATP(p->pagetable);
// return to trampoline.S; satp value in a0.
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was

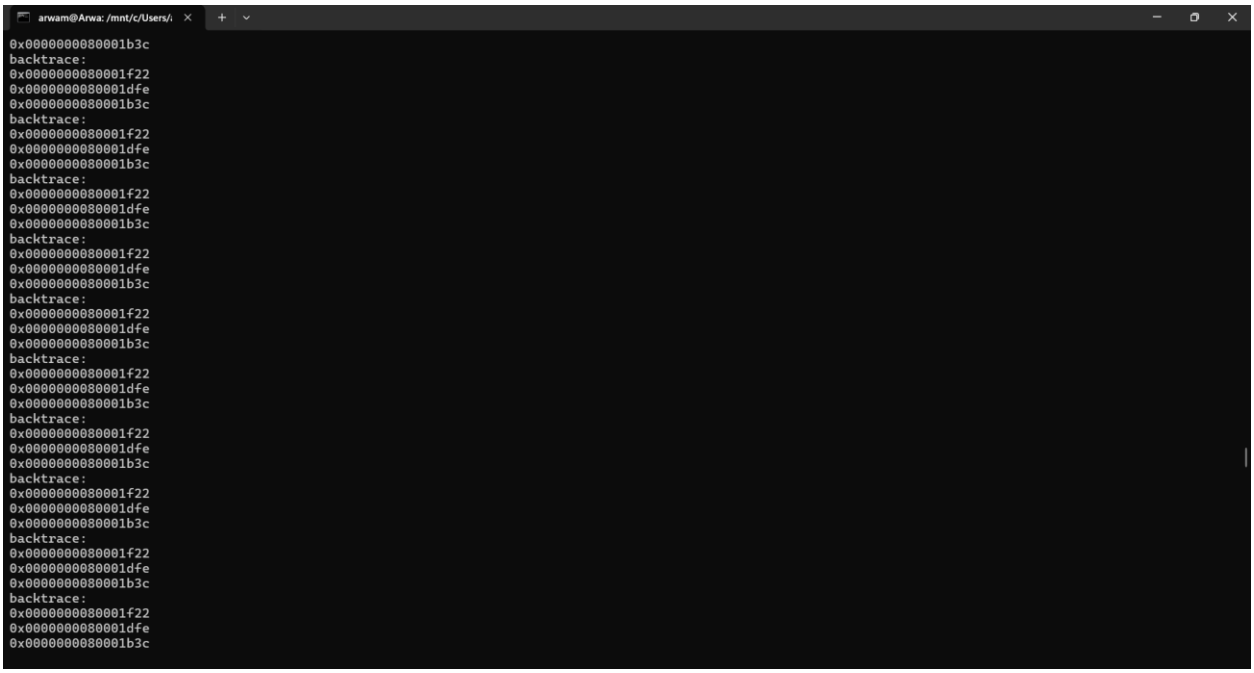

arwam@Arwa: /mnt/c/Users/
GNU nano 7.2
user/user.h
int link(const char *, const char *);
int mkdir(const char *);
int chdir(const char *);
int dup(int);
int getpid(void);
char *sys_sbrk(int, int);
int pause(int);
int uptime(void);
int sync(void);
// ulib.c
int stat(const char *, struct stat *);
char *strcpy(char *, const char *);
void *memmove(void *, const void *, int);
char *strchr(const char *, char c);
int strcmp(const char *, const char *);
char *gets(char *, int max);
uint strlen(const char *);
void *memset(void *, int, uint);
int atoi(const char *);
int memcmp(const void *, const void *, uint);
void *memcpy(void *, const void *, uint);
char *sbrk(int);
char *sbrklazy(int);
// printf.c
void fprintf(int, const char *, ...) __attribute__((format(printf, 2, 3)));
void printf(const char *, ...) __attribute__((format(printf, 1, 2)));
// umalloc.c
void *malloc(uint);
void free(void *);
int sigalarm(int ticks, void (*handler)());
int sigreturn(void);
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was

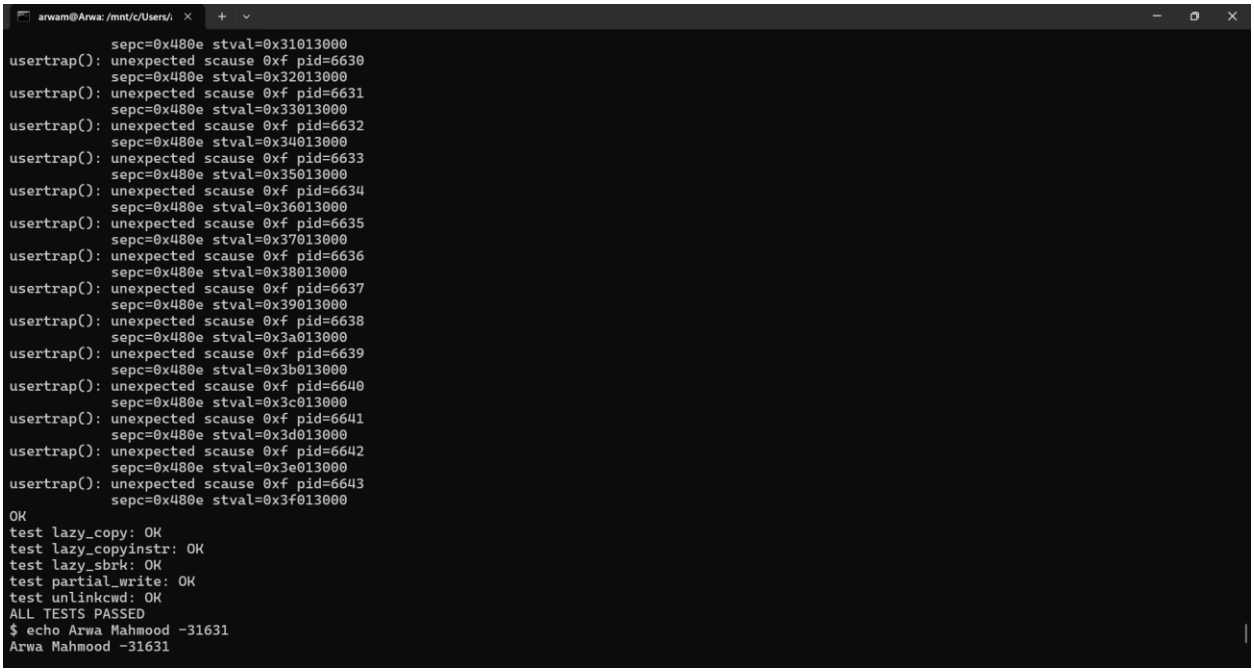

arwam@Arwa: /mnt/c/Users/
GNU nano 7.2
user/usys.pl
print ".global $prefix$name\n";
print "$prefix$name:\n";
} else {
print ".global $name\n";
print "$name:\n";
}
print " li a7, SYS_${name}\n";
print " ecall\n";
print " ret\n";
}
entry("fork");
entry("exit");
entry("wait");
entry("pipe");
entry("read");
entry("write");
entry("close");
entry("kill");
entry("exec");
entry("open");
entry("mknod");
entry("unlink");
entry("fstat");
entry("link");
entry("mkdir");
entry("chdir");
entry("dup");
entry("getpid");
entry("sbrk");
entry("pause");
entry("uptime");
entry("sync");
entry("sigalarm");
entry("sigreturn");
^G Help
^X Exit
^O Write Out
^R Read File
^W Where Is
^\ Replace
^K Cut
^U Paste
^T Execute
^J Justify
^C Location
^/ Go To Line
M-U Undo
M-E Redo
M-A Set Mark
M-6 Copy
M-] To Bracket
^Q Where Was

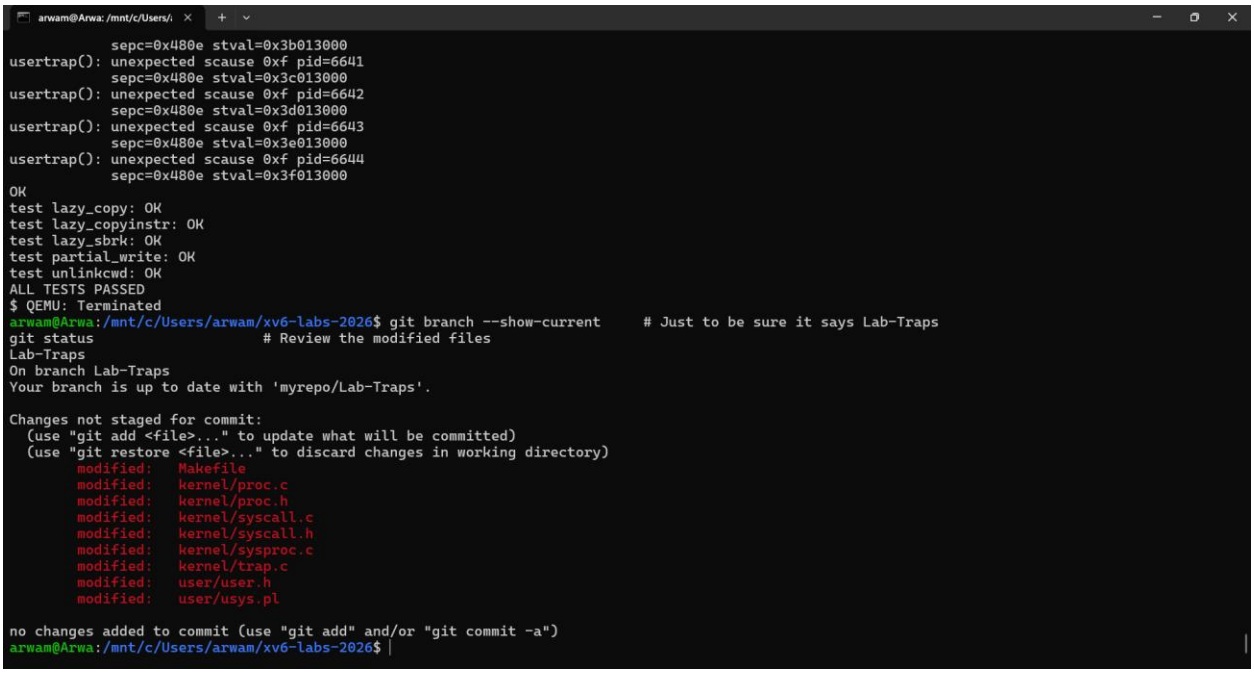

arwam@Arwa: /mnt/c/Users/
usertrap(): unexpected scause 0xf pid=6639
sepc=0x480e stval=0x3b013000
usertrap(): unexpected scause 0xf pid=6640
sepc=0x480e stval=0x3c013000
usertrap(): unexpected scause 0xf pid=6641
sepc=0x480e stval=0x3d013000
usertrap(): unexpected scause 0xf pid=6642
sepc=0x480e stval=0x3e013000
usertrap(): unexpected scause 0xf pid=6643
sepc=0x480e stval=0x3f013000
OK
test lazy_copy: OK
test lazy_copyinstr: OK
test lazy_sbrk: OK
test partial_write: OK
test unlinkcwd: OK
ALL TESTS PASSED
$ echo Arwa Mahmood -31631
Arwa Mahmood -31631
$ alarmtest
test0 start
.....alarm!
test0 passed
test1 start
..alarm!
.alarm!
..alarm!
.alarm!
..alarm!
.alarm!
..alarm!
.alarm!
..alarm!
.alarm!
test1 passed
test2 start
.....alarm!
test2 passed
test3 start
test3 passed
$


arwam@Arwa: /mnt/c/Users/
xv6 kernel is booting
hart 2 starting
hart 1 starting
init: starting sh
$ alarmtest
test0 start
........alarm!
test0 passed
test1 start
..alarm!
.alarm!
..alarm!
.alarm!
..alarm!
.alarm!
..alarm!
.alarm!
.alarm!
..alarm!
test1 passed
test2 start
............alarm!
test2 passed
test3 start
test3 passed
$ echo Arwa Mahmood 31631
Arwa Mahmood 31631
$ usertests -q
usertests starting
test copyin: OK
test copyout: OK
test copyinstr1: OK
test copyinstr2: OK
test copyinstr3: OK
test rwsbrk: OK
test truncate1: OK


arwam@Arwa: /mnt/c/Users/
$ usertests -q
usertests starting
test copyin: OK
test copyout: OK
test copyinstr1: OK
test copyinstr2: OK
test copyinstr3: OK
test rwsbrk: OK
test truncate1: OK
test truncate2: OK
test truncate3: OK
test openiput: backtrace:
0x0000000080001f22
0x0000000080001dfe
0x0000000080001b3c
OK
test exitiput: OK
test iput: OK
test opentest: OK
test writetest: OK
test writebig: OK
test createtest: OK
test dirtest: OK
test exectest: OK
test pipe1: OK
test killstatus: backtrace:
0x0000000080001f22
0x0000000080001dfe
0x0000000080001b3c
backtrace:
0x0000000080001f22
0x0000000080001dfe
0x0000000080001b3c
backtrace:
0x0000000080001f22
0x0000000080001dfe
0x0000000080001b3c
backtrace:
0x0000000080001f22
0x0000000080001dfe
0x0000000080001b3c


arwam@Arwa: /mnt/c/Users/
0x0000000080001b3c
backtrace:
0x0000000080001f22
0x0000000080001dfe
0x0000000080001b3c
backtrace:
0x0000000080001f22
0x0000000080001dfe
0x0000000080001b3c
backtrace:
0x0000000080001f22
0x0000000080001dfe
0x0000000080001b3c
backtrace:
0x0000000080001f22
0x0000000080001dfe
0x0000000080001b3c
backtrace:
0x0000000080001f22
0x0000000080001dfe
0x0000000080001b3c
backtrace:
0x0000000080001f22
0x0000000080001dfe
0x0000000080001b3c
backtrace:
0x0000000080001f22
0x0000000080001dfe
0x0000000080001b3c
backtrace:
0x0000000080001f22
0x0000000080001dfe
0x0000000080001b3c
backtrace:
0x0000000080001f22
0x0000000080001dfe
0x0000000080001b3c


arwam@Arwa: /mnt/c/Users/
sepc=0x480e stval=0x31013000
usertrap(): unexpected scause 0xf pid=6630
sepc=0x480e stval=0x32013000
usertrap(): unexpected scause 0xf pid=6631
sepc=0x480e stval=0x33013000
usertrap(): unexpected scause 0xf pid=6632
sepc=0x480e stval=0x34013000
usertrap(): unexpected scause 0xf pid=6633
sepc=0x480e stval=0x35013000
usertrap(): unexpected scause 0xf pid=6634
sepc=0x480e stval=0x36013000
usertrap(): unexpected scause 0xf pid=6635
sepc=0x480e stval=0x37013000
usertrap(): unexpected scause 0xf pid=6636
sepc=0x480e stval=0x38013000
usertrap(): unexpected scause 0xf pid=6637
sepc=0x480e stval=0x39013000
usertrap(): unexpected scause 0xf pid=6638
sepc=0x480e stval=0x3a013000
usertrap(): unexpected scause 0xf pid=6639
sepc=0x480e stval=0x3b013000
usertrap(): unexpected scause 0xf pid=6640
sepc=0x480e stval=0x3c013000
usertrap(): unexpected scause 0xf pid=6641
sepc=0x480e stval=0x3d013000
usertrap(): unexpected scause 0xf pid=6642
sepc=0x480e stval=0x3e013000
usertrap(): unexpected scause 0xf pid=6643
sepc=0x480e stval=0x3f013000
OK
test lazy_copy: OK
test lazy_copyinstr: OK
test lazy_sbrk: OK
test partial_write: OK
test unlinkcwd: OK
ALL TESTS PASSED
$ echo Arwa Mahmood -31631
Arwa Mahmood -31631


arwam@Arwa: /mnt/c/Users/
sepc=0x480e stval=0x3b013000
usertrap(): unexpected scause 0xf pid=6641
sepc=0x480e stval=0x3c013000
usertrap(): unexpected scause 0xf pid=6642
sepc=0x480e stval=0x3d013000
usertrap(): unexpected scause 0xf pid=6643
sepc=0x480e stval=0x3e013000
usertrap(): unexpected scause 0xf pid=6644
sepc=0x480e stval=0x3f013000
OK
test lazy_copy: OK
test lazy_copyinstr: OK
test lazy_sbrk: OK
test partial_write: OK
test unlinkcwd: OK
ALL TESTS PASSED
$ QEMU: Terminated
arwam@Arwa:/mnt/c/Users/arwam/xv6-labs-2026$ git branch --show-current # Just to be sure it says Lab-Traps
git status # Review the modified files
Lab-Traps
On branch Lab-Traps
Your branch is up to date with 'myrepo/Lab-Traps'.
Changes not staged for commit:
(use "git add <file>..." to update what will be committed)
(use "git restore <file>..." to discard changes in working directory)
modified: Makefile
modified: kernel/proc.c
modified: kernel/proc.h
modified: kernel/syscall.c
modified: kernel/syscall.h
modified: kernel/sysproc.c
modified: kernel/trap.c
modified: user/user.h
modified: user/usys.pl
no changes added to commit (use "git add" and/or "git commit -a")
arwam@Arwa:/mnt/c/Users/arwam/xv6-labs-2026$